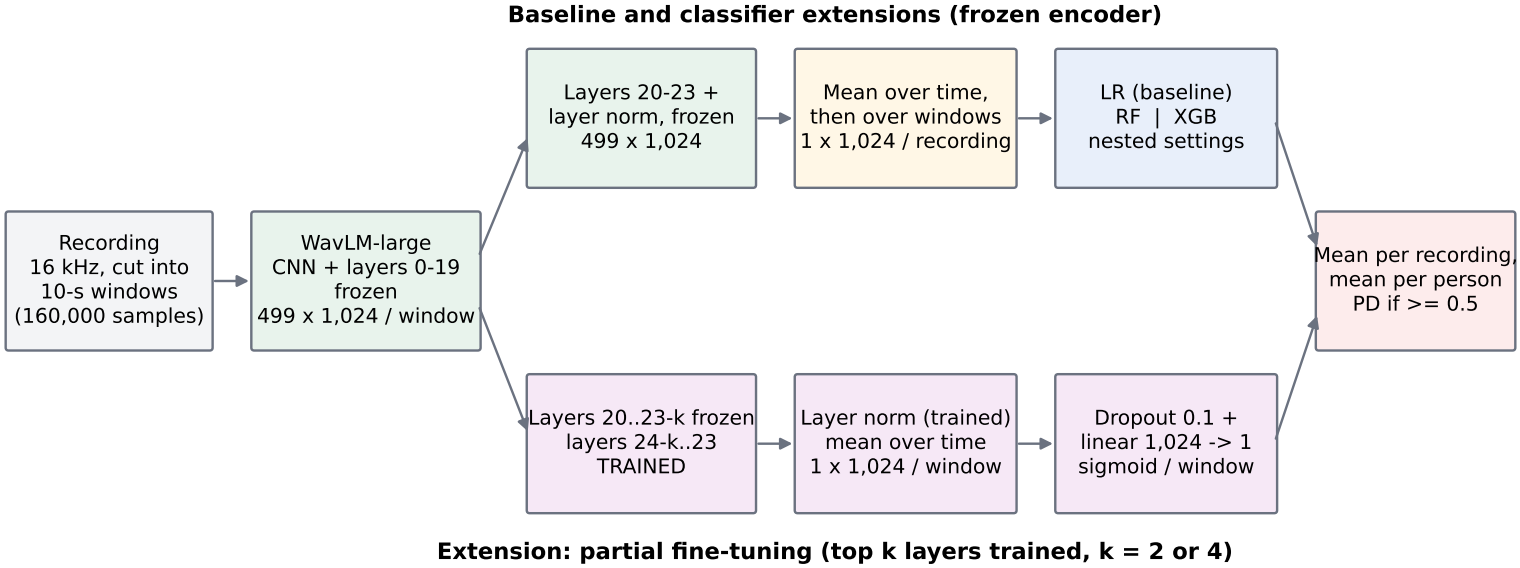
Baseline and classifier extensions (frozen encoder)
Recording
16 kHz, cut into
10-s windows
(160,000 samples)
WavLM-large
CNN + layers 0-19
frozen
499 x 1,024 / window
Layers 20-23 +
layer norm, frozen
499 x 1,024
Mean over time,
then over windows
1 x 1,024 / recording
LR (baseline)
RF | XGB
nested settings
Layers 20..23-k frozen
layers 24-k..23
TRAINED
Layer norm (trained)
mean over time
1 x 1,024 / window
Dropout 0.1 +
linear 1,024 -> 1
sigmoid / window
Mean per recording,
mean per person
PD if >= 0.5
Extension: partial fine-tuning (top k layers trained, k = 2 or 4)